Autonomous High-Speed Low-Altitude Stealth Drone for Urban Navigation

Deep Reinforcement Learning for Stealth Navigation and Object Detection in Urban Airspace

Final Report

Submitted by

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1. Introduction and Background

UAVs have been widely adopted across areas such as logistics, surveillance, and search and rescue. Most autonomous drone systems, however, are built for open airspace, where GPS is reliable, visibility is unobstructed, and obstacles are minimal. Urban environments are fundamentally different: dense buildings, narrow corridors, moving obstacles, and the need to remain undetected in sensitive or contested situations.

This project addresses that gap. We propose an autonomous, high-speed, low-altitude stealth drone system built specifically for urban navigation. Instead of optimising only for speed or shortest path, the drone prioritises routes that use buildings and urban structures as visual cover, reducing its exposure in open areas. The system combines two ML components: a Deep Reinforcement Learning (DRL) agent for real-time stealth navigation, and a fine-tuned object detection model for autonomous target identification. Both are trained and tested in Microsoft AirSim configured with urban cityscape assets. Target applications include autonomous surveillance, defence, and urban reconnaissance.

2. Problem Description and Project Objective

Standard drone navigation systems are not built with stealth as a core constraint. They optimise for speed or shortest path, leaving the drone fully exposed in open airspace. In urban environments, this makes the drone easily detectable by adversaries, surveillance systems, or detection algorithms.

The core challenge this project addresses is: how can a drone autonomously navigate through a dense urban environment at high speed while dynamically choosing routes that minimise detectability, avoid collisions, and still reach a defined target? The drone must also be capable of identifying specific targets using onboard vision alone, without human input. Solving both navigation and detection autonomously, in real time, under strict stealth constraints is a multi-objective ML problem that existing systems do not address.

Project Objective: To develop and validate an autonomous stealth drone system using DRL for low-altitude urban navigation and a fine-tuned object detection model for target identification, integrated within a simulated AirSim urban environment.

3. System Architecture

The system is made up of three integrated layers:

Perception Layer: Receives raw visual frames from AirSim's simulated drone camera and passes them through the fine-tuned object detection model (YOLOv8 / SSD) to identify and localise the target.

Decision Layer: Hosts the DRL navigation agent. It receives the current drone state (position, velocity, heading, obstacle proximity, and estimated visibility exposure) and outputs a flight action. The reward function penalises time in open areas (stealth penalty), collisions (safety penalty), and deviation from the target (goal reward).

Simulation Layer: Microsoft AirSim provides physics-based flight simulation, urban environment rendering, collision detection, and sensor output.

All three layers operate in a closed loop: AirSim provides state observations, the DRL agent acts, AirSim steps forward, and the detection model continuously monitors for the target. Stealth navigation and target detection are tightly coupled throughout the flight.

4. Methodology and Scope

This project uses two ML components working in parallel: a DRL agent for stealth navigation, and a fine-tuned object detection model for target identification. The main tasks are:

1. **Environment Setup:** Configure AirSim with urban cityscape assets.
2. **Data Collection and Preprocessing:** Collect navigation and detection datasets. Preprocessing includes normalization of state observations, image resizing and augmentation (flipping, brightness/contrast variation) for the detection pipeline, and handling any missing sensor values in navigation data.
3. **Baseline Implementation:** Establish baseline models for benchmarking.
4. **DRL Agent Training:** Train a policy network (PPO or SAC) with a custom stealth-aware reward function.
5. **Object Detection Fine-Tuning:** Adapt a pre-trained detection model on drone-perspective data.
6. **System Integration:** Combine both models within the AirSim simulation loop.

Baseline Models

Navigation Baseline (A* / RRT*): These classical pathfinding algorithms serve as the navigation baseline. The DRL agent must demonstrate a superior stealth path compared to these shortest-path algorithms, which have no awareness of cover or detectability.

Detection Baseline (YOLOv8-Nano / SSD): Standard pre-trained YOLOv8-Nano or SSD will serve as the detection baseline. The fine-tuned model must outperform these on drone-perspective target detection. Final choice depends on latency and accuracy trade-offs during experimentation.

Evaluation Metrics

- **Stealth:** Visibility Time (time spent in open, unobstructed areas). Detection Probability may also be incorporated.
- **Navigation:** Success Rate (percentage of episodes reaching the target) and Mean Time to Collision (MTTC). Path Efficiency tracked as a secondary indicator.
- **Detection:** Mean Average Precision (mAP) for accuracy, and Inference Latency given the real-time demands of high-speed flight.

Scope

In-Scope: Training a DRL agent for stealth navigation, fine-tuning an object detection model for target identification, and integrating both into an AirSim urban simulation environment.

Out-of-Scope: Physical drone hardware construction, long-range radio communication protocols, and legal or regulatory permit acquisition.

5. Dataset

Navigation Datasets

- **Zurich Urban MAV Dataset:** Low-altitude urban flight data for realistic navigation training.
- **UAVid Dataset:** High-altitude urban scene dataset, potentially used for contextual scene understanding.

Target Detection Datasets

- **VisDrone Dataset:** A large-scale benchmark captured from drone perspectives, ideal for aerial target identification tasks.
- **COCO Dataset:** A general-purpose object detection dataset providing broad object class variety to improve model generalisation.

6. Tools and Technology

- **Simulation:** Microsoft AirSim (Unreal Engine 4.27) for photorealistic urban flight simulation with physics-based drone dynamics and sensor emulation.
- **Language:** Python 3.10
- **DRL Framework:** Stable-Baselines3 with PyTorch backend (PPO or SAC agents)
- **Object Detection:** Ultralytics YOLOv8 for fine-tuning and inference
- **Data and Training:** NumPy, OpenCV, Albumentations for preprocessing; Weights & Biases for experiment tracking
- **Hardware:** CUDA-enabled GPU (min 8 GB VRAM), Linux or Windows OS with Unreal Engine 4.27

7. Contributions

The workload division below is tentative and may change as the project progresses.

Name	Contribution
Mulham Baig (BSAI23017)	DRL Agent Development: Environment setup (AirSim), reward function design, and navigation policy training.
Safiullah (BSAI23010)	Object Detection Pipeline: Fine-tuning the detection model, dataset preprocessing, and integration with the navigation agent.